Neurosymbolic Verification of LLM-Generated Clinical Discharge and Treatment Plans

Team meeting 1 · Requirements, deliverables, data, plan and risks

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Model A

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Rules and verifier

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Evaluation and baseline

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Infrastructure and verifier

github.com/Akashkumarsenthil/Clinical-Discharge-Treatment-Plan-Verification

REQUIREMENTS

Goals, scope and success criteria

Goal

Check an LLM-written discharge or treatment plan against safety rules the team writes, before a clinician or system acts on it.

In scope for DATA 298A

- Verifier and rule set
- Evaluation harness and benchmark
- Model A: direct LLM plan generation
- Reproduced Hager et al. baseline

Out of scope

- Models B, C and D (DATA 298B)
- Conditions beyond four abdominal pathologies
- Clinical deployment

≥ 0.85 Plan validity rate on 100 held-out cases

≤ 0.05 Constraint-violation escape rate

54.9% Hager et al. baseline: mean diagnostic accuracy (OASST 70B). Different task; not compared with validity

298A is done when Model A runs through the verifier and harness on all 100 test cases, reporting validity, escape rate, the failing rule ID and runtime per plan. Escape rate = share of known-unsafe plans the verifier passes.

DELIVERABLES

Four models, their metrics and artifacts

Model	Purpose	Target metrics	Expected artifacts
A · 298A	LLM writes a plan directly from the case; the floor the others must beat	Measured, not targeted: validity, escape rate and runtime per plan, by pathology	Prompt and generator code, plans for 100 test cases, per-class results
B · 298B	Translate the clinical note and goal into a formal specification	Translation fidelity ≥ 0.90 on a hand-checked validation subset (provisional)	Spec schema, translator, consistency checks
C · 298B	LLM plus Z3 solver: name the broken rule, send the plan back for repair	Validity ≥ 0.85 and escape rate ≤ 0.05 on 100 cases; solver cost per plan reported	Rule set, Z3 verifier, repair loop, counterexample logs
D · 298B	LLM paired with a classical PDDL planner	Validity and escape rate at least Model C's, at comparable solve time (provisional)	PDDL domain, translator, planner run logs

The Hager et al. baseline (54.9% diagnostic accuracy) is a comparison point, not a fifth model. Provisional targets are frozen before each model runs.

LITERATURE SURVEY

Our baseline, and three papers that shape the design

BASELINE WE REPRODUCE

54.9%

Mean diagnostic accuracy, OASST 70B, autonomous setting (Fig. 3)

Hager et al., Nature Medicine 30:2613–2622 (2024)

Problem

Can LLMs make real clinical decisions, not just pass exam questions?

Method

Models gather history, exam, labs and imaging step by step on 2,400 MIMIC-IV admissions, four abdominal conditions

Findings

Worse than physicians; skipped required labs, misread lab ranges, ignored diagnostic and treatment guidelines

How we use it

Same 2,400 cases; we rerun its public code as a separate baseline. It scores diagnosis, we score rule validity

Paper	Problem and method	Result	Relevance to us
LLM+P, Liu et al. 2023	LLM writes PDDL; a planner solves it	Optimal plans for most problems; LLM alone mostly infeasible	Basis for Model D
VeriMed, 2026	LLM formalizes medical-device rules; Z3 counterexamples drive repair	Hemodialysis QA verified accuracy 55.4% to 98.5%	Basis for Model C repair loop
Valmeekam et al. 2023, self-critique	GPT-4 plans checked by GPT-4 itself or by the sound VAL verifier	40/100 base; 55 with LLM check; 88 with VAL	Why the checker is a solver, never the model

RELATED PRODUCTS

What exists today and how we differ

Product	What it offers	Limit for this problem	How Team 6 differs
Amazon Comprehend Medical	Extracts medical entities and links them to ICD-10-CM, RxNorm and SNOMED CT	Reads text; does not judge whether a plan is safe or well sequenced	We test the plan and name the failing rule
Google Vertex AI	Hosts Gemini and other models with grounding, safety filters and evaluation tools	Checks are statistical or model-judged; no guarantee a given rule holds	A solver, not a model, decides pass or fail
OpenAI API	GPT models; structured outputs force schema-valid answers	Schema-valid is not clinically valid; models grade themselves poorly	Z3 checks rules the team wrote
Hospital LLM drafting pilots (Stanford, a Dutch academic hospital)	LLM drafts discharge summaries; 384 discharges at Stanford, Aug to Oct 2025	Safety rests on manual physician review, which is late and does not scale	Automated check runs before physician review

DATA

2,400 admissions, cleaned, split and checked

What we have

- Source: MIMIC-IV-Ext-CDM v1.1 (Hager et al.), an extract of MIMIC-IV v3.1
- Size: 2,400 admissions, 9 tables, 178,747 rows, 18 MB
- Schema: admission note and exam, labs, microbiology, radiology, ICD codes, discharge diagnosis and procedures
- Classes: appendicitis 957, cholecystitis 648, pancreatitis 538, diverticulitis 257
- License: PhysioNet Credentialed Health Data License 1.5.0; signed agreement per person; no patient data in the repo

What we did

- Cleaning: 75 duplicate rows removed; text spacing normalized; lab strings parsed to number and unit, then flagged low, normal, high or unknown with no imputation; outliers kept as real cases
- Splits: 1,840 train, 460 validation, 100 test, stratified by pathology, seed 298, leakage check passed
- Augmentation: none on patient records; synthetic rule-violating plans (rule catalog) supply the known-unsafe cases escape rate needs
- Validation: checksums 12 of 12; 2,400 of 2,400 IDs found in MIMIC-IV; checks pass on clean data and fail on deliberately broken copies

ORGANIZATION PLAN

Roles, tools and workflow

Member	Role
Ritika Mukesh Neema	Model A
Sneha Singh	Rule catalog, verifier rules
Akashkumar Senthilkumar	Data, manifest, pipeline
Rajesh Paruchuri	Metrics, splits, baseline, harness
Shruthi Thirukumaran	Environment, CI, verifier lead

Each work package has one primary owner in Linear; other members contribute through linked issues.

Area	Tool or practice
Code	GitHub repo; every change reviewed by a teammate
Issues	Linear: two-week cycles; one assignee per issue; closed only with a linked artifact
Documentation	Repo README, pipeline notebook, decisions logged in Linear
Experiments	Fixed seed 298, versioned splits and configs; aggregate results only
Data	Stays on each member's machine; never committed
Deployment	None in 298A; local evaluation harness run with one command
Communication	Weekly sync; instructor and ISA feedback becomes Linear issues
Incident response	Patient data committed by mistake: stop, purge history, tell the instructor the same day (proposed)

SCHEDULE

Milestones and dependencies for DATA 298A

Milestone	Depends on	Target
Data pipeline, cleaning and splits	PhysioNet access	Done, Oct 8
Rule set v1, high-severity rules first	Data pipeline	Oct 22
Z3 verifier and evaluation harness	Rule set v1, fixed splits	Nov 5
Model A plans on validation cases	Harness, local LLM setup	Nov 5
Team Meeting 2 demo: Model A through the harness	Model A, harness	Nov 12
Reproduce the Hager et al. baseline	GPU or HPC access, pipeline	By late Nov
100-case test run and failure study	Model A, verifier, harness	Nov to early Dec
Final report and presentation	All of the above	Dec 7 and Dec 8

Checkpoints: Progress Report 2 Oct 22 · Workbook 2 Nov 11 · Team Meeting 2 Nov 12 · Final report Dec 7 · Final presentation Dec 8. Reviews: weekly team sync, pull-request review on every change, instructor and ISA feedback tracked in Linear, demo dry run before each checkpoint.

Risks, mitigations and fallback plans

Risk	Mitigation	Fallback
Imbalance and drift: 3.7 times more appendicitis than diverticulitis; one hospital, 2008 to 2019	Stratified split; report every metric per pathology; claim nothing beyond four conditions	Report 95% intervals; flag the 11 diverticulitis test cases as low-confidence
Missing data: 30.9% of lab rows unflaggable; microbiology in 77% of cases	Keep unknown, never impute; a rule with no input reports not checkable	Report rule coverage next to validity
Operational: no GPU yet for the 70B baseline; Z3 may not scale; repair runs vary	Request university GPU or HPC; high-severity rules first; fixed seeds, repeat runs	Smaller model with the gap documented; ship the verified rule subset
Ethical: patient privacy, single-hospital bias, misuse as an autonomous decision-maker	Data stays local under the license; aggregate outputs only; results by sex and age group; decision support with human review	Incident plan: purge, notify the instructor the same day

STATUS

Where we are and what comes next

Done

- 2,400 cases cleaned, split and leakage-checked
- Workbook 1 submitted; Linear cycles running
- Related work surveyed

Next

- Rule set v1 and Z3 verifier
- Model A plans on validation cases
- Reproduce the Hager et al. baseline

Open items

- GPU or HPC for the 70B baseline
- Which local open-weight model runs Model A
- Secondary-metric targets: repair, fidelity, solver cost

Questions?